The following are the resources I prepared to facilitate a hypothetical energy crisis simulation workshop with U.S. and Australian hydrogen researchers. I incepted and led the development of this workshop with the support of a committee of 3 other students. The event featured 8 delegates from Australia and 8 delegates from the U.S. to collaboratively prepare a response to a hypothetical government response to an energy supply crisis in Indonesia. The workshop helped students develop their strategic thinking skills while also enabling the opportunity for international knowledge sharing amongst delegates. My experience facilitating workshops would be valuable to the role where I could help support cross agency initiatives such as inter-departmental working groups and knowledge sharing to ensure grant policy is aligned with broader government objectives.

Document contents

- Crisis response brief
- Crisis map
- Delegate roles
- Example crisis response solution

## Crisis brief: Stabilising Indonesian seasonal energy storage through hydrogen

Indonesia's recent milestone of 14.5% renewable generation war marred by rolling blackouts, disrupting Indonesian citizens progress to supply the world with critical minerals vital for green energy technologies and halting efforts to increase renewable utilisation.

The archipelago faces power grid instability due to large seasonal swings in renewable energy production. The west coast receives excess solar during the summer but a deficit in winter. Conversely East Indonesia receives surplus wind energy during winter. The net renewable energy generation across the country is sufficient to meet demand yet compartmentalised energy infrastructure has impeded the distribution of renewable energy through the seasons.

The Indonesian Ministry of Energy and Mineral Resources has requested industry and government delegates of Australia and the United States to devise a roadmap for utilising green hydrogen as a seasonal energy storage medium. The roadmap should highlight the specific resources (technology, finances and expertise) that the U.S. and AU will commit to support Indonesia's energy stability while progressing their own interests for clean energy adoption. Delegates should consider the role of green/blue H2 to compliment electrification and the trade-off of temporary vs permanent infrastructure during their response.

Apply the design process (Problem statement – Must/Should - Design paradigms – Down selection) to identify the technology, skills and economic exchanges between countries to restore grid stability in the short (1 year) medium (5 years) and long term (20 years).

| Delegate | Representing stakeholder | Team |
|---|---|---|
| | US Government | 1 |
| | AU Government | |
| | US Industry | |
| | AU Industry | |
| | AU Government | 2 |
| | AU Industry | |
| | US Industry | |
| | US Government | |
| | AU Government | 3 |
| | US Industry | |
| | AU Industry | |
| | US Government | |
| | AU Government | 4 |
| | US Industry | |
| | AU Industry | |
| | US Government | |

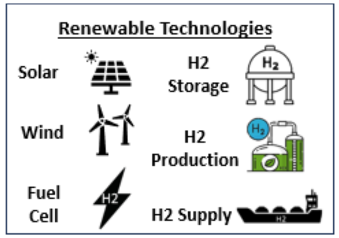
**Crisis response agenda**

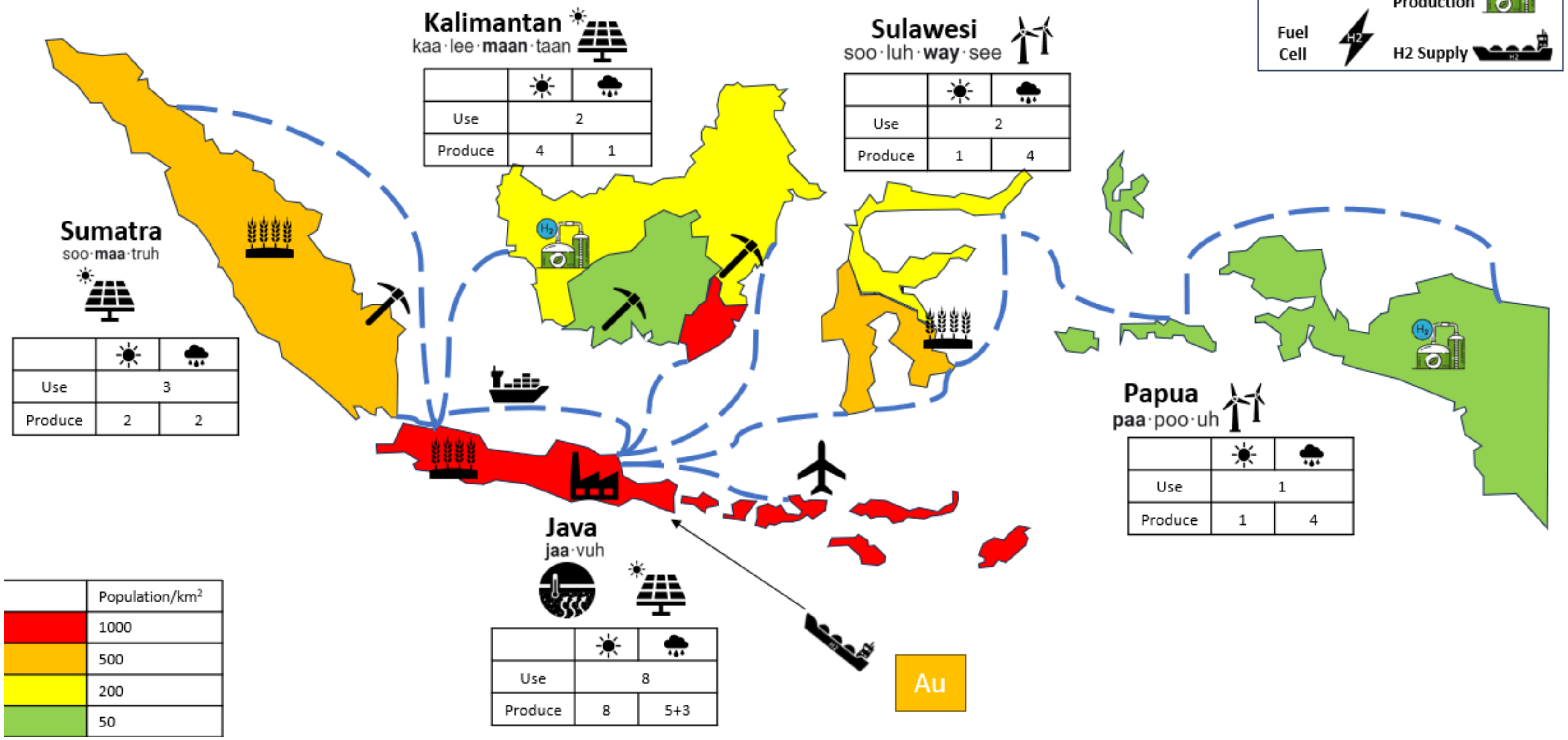
| Time AU | Time US | Activity |
|---|---|---|
| 10:00 | 3:00 | Event commencement |
| 10:05 | 3:05 | Crisis briefing |
| 10:15 | 3:15 | Commence crisis response as teams |
| 11:20 | 4:20 | Reconvene to share team strategies and assumptions through QnA |
| 11:50 | 4:50 | Final thoughts |
| 12:00 | 5:00 | Event finish |

## Crisis problem statement map

- Units of power are in GW
- Data listed is renewable energy, comprising 30% of Indonesians electricity demand
- 1 GW electrolysis = 450 Tonne/day H2

# 2024-2025




Kalimantan

| | | |
|---|---|---|
| Use | 2 | |
| Produce | 4 | 1 |

Sulawesi

| | | |
|---|---|---|
| Use | 2 | |
| Produce | 1 | 4 |

Sumatra

| | | |
|---|---|---|
| Use | 3 | |
| Produce | 2 | 2 |

Papua

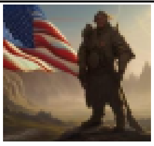
| | | |
|---|---|---|
| Use | 1 | |
| Produce | 1 | 4 |

Java

| | | |
|---|---|---|
| Use | 8 | |
| Produce | 8 | 5+3 |

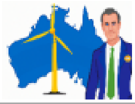
| | Population/km$^2$ |
|---|---|
| | 1000 |
| | 500 |
| | 200 |
| | 50 |

## Role Overviews

| | |
|---|---|
|  | |
| **Actor:** | US Government |
| **Interests and Goals:** | • Interest in supporting US hydrogen technology development in preparation for domestic H2 hubs opportunity.<br>• Foster international hydrogen supply chain, including manufacturing.<br>• Expand economic and strategic ties with Indonesia and ASEAN Countries.<br>• Support jobs, including legacy jobs in the fossil fuel industry. |
| **Capabilities include (but are not limited to):** | • Can purchase 2 clean energy (CE) technologies per year.<br>• (GP) Can invest in a hydrogen technology center which will produce 5 energy units per year autonomously after 7 years of support from industry personnel.<br>• (CE) Can supply 1 mobile hydrogen liquefier that can supply 2 clean energy units per year.<br>• Can raise taxes on carbon-intensive industries.<br>• Able to borrow relatively cheaply on international markets.<br>• Ability to print money. |

| | |
|---|---|
|  | |
| **Actor:** | US Industry |
| **Interests and Goals:** | US "Clean Energy" Technology Companies (ie: manufacturers of electrolysers, batteries, turbines & engines, fuel cells)<br>• Demonstrate hydrogen technologies for prospective customers (both domestic and international), while retaining IP.<br>• Expand customer base.<br>• Gain access to low-cost manufacturing base.<br><br>US Energy Companies (Oil & gas)<br>• Interest in demonstrating hydrogen supply chain and "green credentials".<br>• Preference toward blue hydrogen and carbon capture.<br>• Relatively risk-averse - unwilling to take steps without funding support. |
| **Capabilities include (but are not limited to):** | • Can supply personnel to operate 1 government project (GP) at a time.<br>• (CE) Can sell 1 wind mill per year for installation on 1 island.<br>• Multinational companies can pull expertise from Europe and Asia<br>• Can raise capital or acquire technology start-ups |

| |  |
|---|---|
| **Actor:** | Australian Government |
| **Interests and Goals:** | • Interest in establishing an international hydrogen export chain and relationships for long term hydrogen export.<br>• Maintain and expand economies ties with Indonesia and ASEAN Countries.<br>• Support legacy jobs in the fossil fuel industry in the short term. |
| **Capabilities include (but are not limited to):** | • (GP) Can supply 1 energy unit per year to any island. This can be done for a maximum of 3 years by the government but then must be done with industry personnel.<br>• (GP) Can fund a hydrogen liquefier that produces 2 energy units per year on one island but this must be done with AU Industry personnel and they must work for 3 years prior to the system coming online.<br>• Can purchase 1 clean energy (CE) technology per year.<br>• Can approve land usage for renewable projects in Australia<br>• Ability to print money. |

| |  |
|---|---|
| **Actor:** | Australian Industry (Oil & Gas, mining, steelmaking) |
| **Interests and Goals:** | • Interest in demonstrating hydrogen supply chain and "green credentials".<br>• Signing contracts for long-term hydrogen export.<br>• Relatively risk-averse - unwilling to take steps without funding support.<br>• "Agnostic" toward hydrogen, ammonia, e-methanol, e-methane and other e-fuels. Likely to have preference toward blue hydrogen and carbon capture. |
| **Capabilities include (but are not limited to):** | • Can supply personnel to operate 1 government project (GP) at a time.<br>• (CE) Can sell 3 hydrogen transporters that ferry energy from 1 island to another, each operating for 1 year.<br>• (CE) Can sell 1 solar panel for 1 island.<br>• Can leverage existing LNG workforce for cryogenic hydrogen. |

For every 2 excess energy units produced, 1 additional clean energy (CE) technology can be purchased each year.

## Crisis response solution

<table>
<tr>
<td>Problem consolidation<br><br>Finish time<br>10:27 AU | 3:27 US</td>
<td colspan="2">Observations of Indonesia’s energy mix / how H2 could compliment electrification / other energy vectors
<ul>
<li>Energy supply = demand, but not where it is needed</li>
<li>Larger populations consume more energy</li>
<li>Different industries may need different energy storage</li>
<li>Logistics paths could be redefined</li>
<li></li>
</ul>
</td>
</tr>
<tr>
<td>Crisis response criteria<br><br>Finish time<br>10:35 AU | 3:35 US</td>
<td>Strategy must:
<ul>
<li>Enhancement – better than before situation</li>
<li>Meet all energy requirements throughout year</li>
<li>Match energy demand and use in each area</li>
</ul>
</td>
<td>Strategy should:
<ul>
<li>Build relations for clean energy supply chain across ID, AU and US</li>
<li>Making use of available land in less populated areas for energy production/storage</li>
<li>Utilise existing energy infrastructure to accelerate renewables</li>
</ul>
</td>
</tr>
<tr>
<td colspan="3"></td>
</tr>
<tr>
<td>Short term strategy<br>1 year<br><br>Finish time<br>10:50 AU | 3:50 US</td>
<td colspan="2">Paradigms
<ul>
<li>Diversified renewable energy production for rural communities</li>
<li>Transport renewable energy via ship or cable to where it is needed</li>
<li>Store summer energy for winter via H2 in each island</li>
<li></li>
</ul>
Response down selection
<table>
<tr><td>Activity</td><td>Responsible stakeholder</td></tr>
<tr><td>Supply temporary infrastructure for energy storage</td><td>US Industry</td></tr>
<tr><td>Provide renewable power tech</td><td>AU Industry</td></tr>
<tr><td>Grant/loan to support tech deployment</td><td>AU US govt</td></tr>
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<tr>
<td>Medium term strategy<br>5 years<br><br>Finish time<br>11:05 AU | 4:05 US</td>
<td colspan="2">Paradigms
<ul>
<li>Transport renewable energy via ship or cable to where it is needed</li>
<li>Liquid hydrogen storage for energy independence in each island</li>
<li>Diversified renewables in each area incl. hydro</li>
<li></li>
</ul>
Response down selection
<table>
<tr><td>Activity</td><td>Responsible stakeholder</td></tr>
<tr><td>Material support to build new renewables and LH2</td><td>US industry</td></tr>
<tr><td>Leverage LNG expertise to help build energy storage</td><td>AU industry</td></tr>
<tr><td>Further loan support for renewables investment</td><td>US AU govt</td></tr>
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<tr><td colspan="2"></td></tr>
<tr><td>Long term strategy<br>20 years<br><br>Finish time<br>11:20 AU | 4:20 US</td><td>Paradigms
<ul>
<li>Utilize energy storage/independency to support new industries and cities in less populated areas</li>
<li>Large energy storage capacity to utilise Indonesia as a pacific battery</li>
<li>Selling dirty energy assets or converting it to mining for funds to support further renewable adoption</li>
<li>Expansion of renewable capacity for affordable path to 100% renewables</li>
</ul>
Response down selection
<table>
<tr><td>Activity</td><td>Responsible stakeholder</td></tr>
<tr><td>Discount on Indonesia export to AU and US in return for support for energy independence</td><td>ID govt</td></tr>
<tr><td>Financial and material support for large renewable projects – in return for energy imports to US</td><td>US AU Industry</td></tr>
<tr><td>Financial support for HVDC cable for energy storage across ID and AU</td><td>AU Govt</td></tr>
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